Historical abstract loss and use-dependent robustness of a semantic novelty indicator

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Abstract

Historical abstract availability can affect different uses of the same semantic novelty indicator. We study an established nearest-prior semantic-distance score using SPECTER, simulating historical abstract loss by substituting authentic titles while fixing focal texts and retaining every historical record. The experiments include 1,557 focal computational-linguistics conference papers and 1,471 focal PLOS Computational Biology and PLOS Genetics papers, with 200 assignments per condition. Under random 25% loss, source-contrast distortion averaged 0.0175 and 0.0167 baseline score standard deviations (SD), while 14.10% and 14.90% of baseline top-decile selections were replaced. Post hoc descriptive analyses show that these small contrast changes coexist with whole-cohort mean score increases of 0.1094 and 0.1116 SD. Source preference reduced replacement relative to matched controls in the conference collection but increased it in the biomedical collection. A separately specified post hoc comparison of the same 25% masks found more replacement under record removal than title fallback in all tested conditions, without uniform improvement across all readouts. A limited SPECTER2 sensitivity analysis repeated selected within-encoder masking patterns, but complete-history top-decile overlap between encoders was only 51.9% and 41.9%. These conditional results distinguish perturbation sensitivity, score drift and representation agreement; they do not establish scientific originality, validated selection costs or general policy superiority.

Keywords: semantic novelty, abstract coverage, measurement validity, ranking stability, missing metadata, scientometrics

Classification codes: MSC 2020: 62P25; JEL: C15 , C18

1 Introduction

A claim that a scientometric indicator is robust is incomplete without specifying what the indicator is used to measure. A single score can compare publication groups, order individual papers, or select a high-scoring subset. These uses retain different information from the same score vector. Small changes in a group comparison can coexist with changes in ordering and selection, while selection changes can be concentrated among papers close to a cutoff. Evaluating one output therefore leaves the dependability of the other uses unresolved. For example, a hypothetical evaluator could compare two publication sources by mean score and use the same indicator to shortlist high-scoring papers. A source comparison could change little even when the shortlist changes. This is an illustration of distinct measurement uses, not a claim about an institution's practice or validated selection costs.

For reference-dependent semantic novelty indicators, historical metadata are part of the measurement procedure. The nearest-prior construction compares a focal paper with its most similar earlier publication (Arts et al., 2025). If historical abstracts become unavailable, retaining the records through a title-only fallback changes their representations even though the focal papers and number of reference opportunities remain fixed. The result depends on how incomplete historical text is handled, which records are affected, and which output is examined. A coverage percentage alone does not specify those conditions.

Prior research has established semantic novelty constructions, variation across text inputs and bibliometric uncertainty (Donner, 2024; Shibayama et al., 2021). The closest reference-pool benchmark examines controlled redundancy, relevance and temporality, rank displacement, and differences between aggregate trends and individual consistency (Liu & Zhai, 2026). The unresolved question addressed here is more specific: when authentic titles replace selected historical title-plus-abstract inputs, how do a predefined source contrast, whole-cohort ranking and focal high-score selection respond together, and how does that response depend on missingness structure?

We examine this question for an established nearest-prior score under a pinned SPECTER representation (Arts et al., 2025; Cohan et al., 2020). The intervention preserves focal text, historical record count and strictly earlier comparison years. Structured assignments are compared with random controls preserving specified historical opportunity counts. Cutoff diagnostics then distinguish exchanges among nearby papers from losses extending beyond a fixed selection neighborhood and positive baseline score margins. The joint evaluation measures when small source-contrast distortion on the between-paper SD scale coexists with changes in ranking and selection, and locates limits to that pattern across mechanisms, collections and representations.

Three questions organize the original experiments. **RQ1:** Under historical abstract-to-title substitution, how do absolute source-contrast distortion, whole-cohort rank displacement and replacement of baseline top-decile selections differ? **RQ2:** How do source-preferred and lexical-proxy-preferred assignments differ from random controls with the same historical-year counts, and additionally the same year-by-source counts for the ACL lexical comparison? **RQ3:** Which random and

source-preferred responses recur, and which change, in a separately acquired second collection under the same score construction?

The study includes Association for Computational Linguistics (ACL) and Empirical Methods in Natural Language Processing (EMNLP) conference papers and a biomedical collection, with each original protocol locally frozen before its own scoring outcomes. The second collection follows the first study and provides a qualified replication. Separately labelled post-confirmation checks examine current database coverage, interval sensitivity and three masking conditions under the related SPECTER2 representation. The last check also exposes disagreement in the complete-history biomedical source contrast. Together, the analyses distinguish stability under a specified historical-text perturbation, differences between measurement uses, and dependence of the baseline comparison on representation choice. Agreement on the first does not establish agreement or validity on the others.

2 Related work

Semantic novelty measures operationalize relations between publications and prior knowledge; their meaning depends on the comparison object, text input and representation. Arts et al. (2025) define Semantic Distance as one minus the maximum SPECTER cosine similarity to publications from the preceding five years. We retain that nearest-prior construction with a declared three-year history. Their title-only validation addresses other text metrics and excludes Semantic Distance. Our two-source, 2018–2020 reference frame is narrower than that five-year implementation; all conclusions condition on this declared history rather than a complete scientific literature. Shibayama et al. (2021) instead measure semantic distances among a paper's cited documents, comparing title, abstract and keyword inputs. These established approaches motivate explicit specification of what is represented and compared, rather than treating semantic novelty as a unique measurement procedure.

Earlier work also measures element novelty using minimum or low-quantile cosine distances from a document's title-and-abstract representation to earlier documents (Yin et al., 2023). We therefore treat the nearest-prior construction as established and examine its response to a specified historical-text handling policy. A correction by The PLOS One Staff (2026) restores production-omitted cells in Table 4 of Shibayama et al. (2021); it is not a retraction. We do not use that table's numerical validation estimates here.

Representation and retrieval choices already receive substantial attention. SPECTER supplies citation-informed document representations (Cohan et al., 2020). Kruzenshtern et al. (2026) evaluate token-loss novelty features across domains and against paraphrased focal abstracts. Shen et al. (2026) compare title, summary and combined inputs for nearest-neighbor novelty baselines and assess agreement with expert ratings. Their summary input is not assumed equivalent to an authentic abstract. Shahid et al. (2025) examine novelty assessment alongside changes in retrieved-document membership and ordering after retrieval and reranking ablations. The selected objects there are supporting papers, whereas our selection endpoint concerns the focal publication cohort. Such precedents establish the relevance of

representation choices and selection sensitivity without making these evaluations interchangeable.

Reference collections are also an established part of novelty measurement. Liu and Zhai (2026), in their version 2 preprint, test redundancy, relevance and temporality through controlled pool changes, including equal-count temporal windows, and assess rank displacement in an adapted acceptance tournament. They explicitly distinguish aggregate score trends from individual consistency. Their benchmark is the closest methodological precedent: neither pool sensitivity nor divergence between aggregate and individual behavior originates here. Wang et al. (2026) compare multiple abstract encoders, evaluate approximate-neighbor fidelity and neighborhood-size sensitivity, and analyze journal-network stability. Their novelty score is one minus the mean cosine similarity to ten earlier neighbors and supports citation-group and journal-quartile comparisons. SemNovel likewise applies a biomedical semantic-distance measure to citation validation, prize-related comparisons and research trajectories (Peng et al., 2025). These studies show that semantic indicators support multiple analytical uses.

Metadata quality adds a distinct source of measurement uncertainty. Delgado-Quirós and Ortega (2024) document differences in scholarly metadata completeness; Kim et al. (2026), a preprint, examine integrity problems among records with an abstract field. Availability therefore differs from accurate textual content. Donner (2024) estimates bibliometric data errors and propagates uncertainty into indicators, while perturbation-based ranking and subset-overlap assessments have a longer scientometric history (Freyer, 2014). Database-wide coverage statistics, however, cannot determine omission frequencies or indicator effects within a selected complete-text frame.

Against these precedents, we evaluate a specific historical-information policy jointly for three uses of one indicator. Authentic title substitution retains every historical record and leaves focal representations fixed; the same assignments yield source-contrast distortion, whole-cohort rank displacement and top-decile replacement. Opportunity-matched controls, cutoff qualification and a second collection distinguish conditional response patterns and their limits of transport. A bounded related-encoder check then separates sensitivity to historical information within a representation from agreement of the baseline comparison across representations. The contribution is this controlled measurement-use assessment and its implications for validation, rather than a new score or a general discovery that incomplete information changes rankings.

3 Materials and methods

3.1 Study sequence and collections

An exploratory ACL/EMNLP pilot informed the endpoint choice and precision planning. The 2021 ACL/EMNLP confirmation protocol was locally frozen before its scoring outcomes. After those results, a separately acquired biomedical experiment was specified and frozen before its own outcomes. This sequence provides a qualified

second-collection test using shared representations and analysis, rather than externally preregistered or fully independent validation. The unsuccessful pilot diagnostic is retained in Supplementary Section S7.

ACL/EMNLP comprises all 1,557 eligible 2021 main-conference long/short research papers (710 ACL; 847 EMNLP), evaluated against 3,799 eligible 2018–2020 papers. Official Anthology metadata and deterministic identifier, DOI and exact-text checks define the frame. One historical duplicate was removed. Findings, workshops, demonstrations, tutorials and front matter are outside the frame.

The biomedical frame joins PLOS Computational Biology and PLOS Genetics full Research Articles to PubMed by exact DOI. Eligibility requires complete title/abstract text and a valid electronic publication date agreeing in year with the publisher date. Type, retraction and expression-of-concern rules exclude 34 distinct papers from 4,896 candidates. The retained frame contains 3,391 historical and 1,471 focal papers, including 934 Computational Biology and 537 Genetics focal papers. Supplementary Section S1 reports all exclusions and coverage denominators. Both studies use selected current complete-text frames with 2018–2020 history and 2021 focal papers. The ACL/EMNLP protocol selected 2021 as the earliest untouched cohort after the 2020 pilot. The analysis does not evaluate alternative historical windows or estimate robustness for literature outside these sources.

3.2 Score and historical title substitution

We evaluate the established nearest-prior semantic-distance construction (Arts et al., 2025) using pinned SPECTER representations (Cohan et al., 2020). Title-plus-abstract and authentic title-only inputs share the tokenizer, final-layer CLS pooling, L2 normalization and 512-token cap. Here *full* means title plus abstract. The cap truncates 67 biomedical full inputs (43 historical; 24 focal), no ACL full inputs and no title inputs. Model identities and implementation settings are given in Supplementary Sections S3 and S8.

Let F denote the focal set, H the strictly earlier historical set, f_i a focal full vector, h_j a historical full vector and t_j its title-only counterpart. Baseline and masked scores are

$$b_i = 1 - \max_{j \in H} \langle f_i, h_j \rangle, \qquad z_{im} = 1 - \max_{j \in H} \langle f_i, (1 - M_{jm}) h_j + M_{jm} t_j \rangle. \tag{1}$$

The binary mask M_{jm} substitutes authentic title-only vectors for selected historical papers. Every focal representation and historical paper remains in place (Fig. 1). Exact similarities determine the maximum after temporal/self exclusions. The intervention models unavailable abstracts under a title-only fallback; it neither deletes papers nor imputes abstracts. Title substitution can change a score in either direction.

3.3 Masking and matched controls

ACL rates are 10%, 25% and 50%; biomedical rates are 25% and 50%. These are synthetic scenarios, with 50% a stress condition. Each assignment substitutes exactly $\lfloor p|H| + 0.5 \rfloor$ historical representations. Uniform masks sample without replacement. Source-preferred masks assign exponential sampling rates 8 to ACL or Computational Biology and 1 to the other source, selecting the earliest clocks. ACL additionally

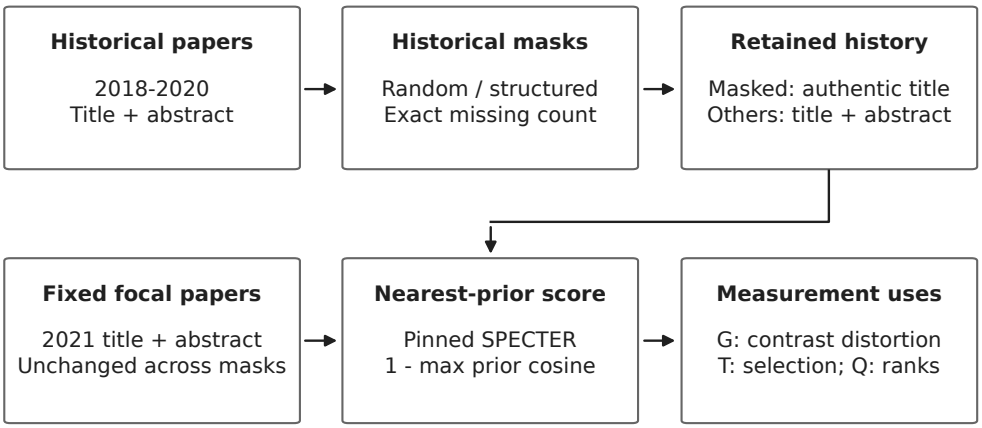


Fig. 1 Controlled historical title substitution. Fixed focal title-plus-abstract inputs are compared with retained historical papers. Selected historical full representations become title-only representations. Each mask generates the same three endpoints, and structured masks have opportunity-matched random controls

tests an 8:1 preference for a fixed translation-title-keyword proxy. This proxy is not a validated topic classification; 8:1 sampling rates are not exact marginal inclusion probabilities or odds. Additional descriptive summaries in Supplementary Section S13.4 report realized loss fractions within the preferred and nonpreferred strata, separately from each stratum's share of the masked set.

Every structured mask has an independently seeded uniform control preserving its realized missing count by historical year. ACL lexical controls also preserve source within year. These controls fix specified opportunity counts while retaining differences in affected content; they do not isolate source-label causation. Each condition has 200 complete assignments, fixed before outcomes: 1,800 base masks plus 1,200 controls in ACL, and 800 plus 400 in the biomedical study. Supplementary Sections S2–S3 give protocol chronology, seeds and algorithms.

3.4 Measurement uses and inference

For focal sources A, B, let s_0 be the population SD of all baseline focal scores. The endpoints are

$$G_m = \frac{|(\bar{z}_{A,m} - \bar{z}_{B,m}) - (\bar{b}_A - \bar{b}_B)|}{s_0}, \tag{2}$$

$$T_m = \frac{|S_0 \setminus S_m|}{k}, \qquad Q_m = \frac{1}{|F|} \sum_{i \in F} \frac{|r_{im} - r_{i0}|}{|F| - 1}. \tag{3}$$

G is absolute distortion of one source contrast, calculated before averaging over masks. S_0, S_m contain the highest $k = \lceil .10|F| \rceil$ scores ($k = 156$ or 148); lexical IDs resolve exact membership ties. T is replacement among baseline selections, accompanied by

expected paper counts. Q is average absolute whole-cohort rank displacement, using average ranks for exact score ties.

Complete masks, or structured/control pairs, are the units of inference. Paired differences are formed before means and variances. The ACL family contains 27 base and 18 paired means; the separate biomedical family contains 12 base and six paired means. Approximate simultaneous intervals are

$$\bar{x} \pm t_{199,\,1-.05/(2J)} s_x/\sqrt{200}, \qquad J \in \{45, 18\}. \tag{4}$$

Bonferroni permits dependence among endpoints, while Student coverage relies on the mean approximation for bounded or skewed outcomes. Intervals concern assignment expectations conditional on the fixed corpus, encoder and mechanism. Supplementary Sections S4–S5 distinguish mean uncertainty, assignment variability and descriptive same-mask frequencies.

Operating tolerances are $G = .10$ baseline SD, $T = .05$ of baseline selections and $Q = .01$ of the rank range. They are study choices, without externally validated acceptable-loss interpretations. The primary joint rule requires $U_G < .10$ and $L_T > .05$; rank displacement is assessed separately. A prespecified three-tolerance grid reuses these intervals. Continuous estimates, source-contrast scales and expected paper counts guide interpretation.

3.5 Additional descriptive analyses

Additional descriptive analyses were specified after inspection of the original experimental outcomes and are not part of the original simultaneous inference families. They summarize the stored results and examine baseline cross-encoder agreement and nearest-neighbor behavior; their scope and chronology are documented in Supplementary Section S13. Write $\Delta_0 = \bar{b}_A - \bar{b}_B$ and $w = n_A/n$. Alongside the primary mean absolute distortion, we report the signed masked contrast $C_m = (\bar{z}_{A,m} - \bar{z}_{B,m})/s_0$, its signed change $D_m = C_m - \Delta_0/s_0$, and whole-cohort mean drift $M_m = \overline{z_m - b}/s_0$. Source-specific shifts $A_m = (\bar{z}_{A,m} - \bar{b}_A)/s_0$ and $B_m = (\bar{z}_{B,m} - \bar{b}_B)/s_0$ obey $D_m = A_m - B_m$ and $M_m = wA_m + (1-w)B_m$. Thus $A_m = M_m + (1-w)D_m$ and $B_m = M_m - wD_m$. These identities also hold for mask means; they do not recover covariance or per-mask uncertainty from marginal summaries. In general, $\overline{G_m} = \overline{|D_m|} \neq |\overline{D_m}|$, and paired ΔG is a difference between absolute distortions, not a signed source-contrast change.

A separately specified post hoc policy comparison reused the same 200 original 25% random masks, 200 source-preferred masks and 200 original year-matched control masks in each collection. Using the saved SPECTER similarities, it compared title fallback with removal of those historical records while fixing focal papers, complete-history baselines and selection rules. Removal deliberately reduces candidate count and is an alternative handling policy, not a count-preserving control. Supplementary Section S14 reports all six collection-by-condition comparisons, descriptive assignment variability and a baseline-score selection-shortfall measure; it adds no tests or inferential family.

Table 1 Fixed complete-text collections and baseline measurement scales.

Collection	History	Focal	k	Contrast/SD
ACL/EMNLP	3,799	1,557	156	−0.130701
Biomedical	3,391	1,471	148	+0.261287

History: 2018–2020; focal: 2021. Source order is ACL minus EMNLP and Computational Biology minus Genetics. Adjacent cutoff gaps are 0.000896216 and 0.00402522 baseline SD. Neither boundary has exact ties. Full cohort accounting is in Supplementary Table S1.

3.6 Cutoff diagnostics and computational verification

The baseline cutoff is the midpoint between the kth and $(k + 1)$th descending scores. A fixed 9th–11th-percentile rank band, expanded through boundary ties, identifies immediate cutoff neighbors. Losses inside and outside it use the original k denominator. Separate diagnostics count losses whose positive baseline score margin exceeds $.01, .025, .05$ or $.10s_0$. Rank-band and score-margin criteria overlap and are not additive. Whole-cohort movement distributions complement Q (Supplementary Section S6).

Restoration identities, numerical checks using separate implementations and isolated scoring/readout reconstruction from frozen embeddings support computational verification. Supplementary Section S8 distinguishes this scope from fresh raw-to-encoder reproduction.

3.7 AI assistance

ChatGPT and OpenAI Codex (OpenAI) assisted with code development, refinement and checking; manuscript organization and revision; and grammar correction and wording refinement. This support included suggestions for descriptive analyses and interpretation considered during revision, and implementation of author-approved analyses. The research idea and original manuscript draft originated with the author, who reviews the code and manuscript and retains responsibility for the methods, results, interpretation and final text. Exact served model builds were unavailable.

4 Results

4.1 Baseline measurement scales

The original SPECTER source contrasts were $-0.130701s_0$ for ACL minus EMNLP and $+0.261287s_0$ for Computational Biology minus Genetics (Table 1). Consequently, the operational $0.10s_0$ tolerance is about 77% and 38% of their absolute magnitudes. Low distortion on the between-paper SD scale therefore does not imply negligible change relative to the source contrast itself. Neither baseline selection boundary had exact ties, but the adjacent-score gaps were small.

4.2 Source-contrast, selection and ranking responses

The same historical intervention produced different responses across measurement uses (Fig. 2). Across all nine ACL conditions, mean source-contrast distortion ranged from 0.0069 to 0.0471 baseline SD, with every simultaneous upper bound below 0.10 SD. Replacement nevertheless ranged from 4.59% to 27.22% of the 156 baseline selections; mean rank displacement ranged from 2.04% to 11.00% of the rank range. Under random 25% loss, distortion was 0.0175 SD [0.0151, 0.0199], replacement was 14.10% [13.56, 14.63] (22.0 expected papers), and rank displacement was 5.18%.

Random biomedical loss reproduced this separation. At 25%, distortion was 0.0167 SD [0.0139, 0.0194], replacement 14.90% [14.42, 15.39] (22.1 of 148 papers), and rank displacement 5.41%. At 50%, mean distortion was 0.0267 SD and replacement 25.55% (37.8 papers). Source-preferred biomedical loss supplied a boundary: distortion reached 0.1486 SD [0.1452, 0.1520] at 25% and 0.3770 SD [0.3715, 0.3826] at 50%, alongside replacement of 17.91% and 32.03%. This qualified replication preserves the random-loss pattern while showing that its small-distortion component does not transport uniformly to structured loss.

Signed contrasts and score levels.

Additional descriptive summaries clarify what the small random-loss contrasts omit (Table 2). At random 25% loss, mean scores rose by 0.1094 SD in ACL and 0.1116 SD in the biomedical collection. The corresponding source shifts were +0.1180/+0.1021 SD (ACL/EMNLP) and +0.1076/+0.1185 SD (Computational Biology/Genetics). Thus both source means shifted upward while their difference changed little. Biomedical source preference instead amplified the positive signed contrast from +0.2613 to +0.4099 SD at 25% and +0.6383 SD at 50%, increases of 56.9% and 144.3% relative to its baseline magnitude. The source shifts were +0.1842/+0.0356 SD and +0.4735/+0.0965 SD. These are changes in the defined score, not evidence of increased scientific merit. Full base and control summaries are in Supplementary Section S13.1.

Table 2 Additional descriptive contrasts and mean drift in four specified conditions. All quantities use the original collection-specific s_0. C_0 and $\overline{C}$ are signed baseline and masked source contrasts; $\overline{G}$ averages absolute distortion; $\overline{M}$ is whole-cohort drift. These means have no new inferential intervals

Collection	Condition	C_0	$\overline{C}$	$\overline{G}$	$\overline{M}$
ACL	Random 25%	-0.131	-0.115	0.018	+0.109
Biomedical	Random 25%	+0.261	+0.250	0.017	+0.112
Biomedical	Source 25%	+0.261	+0.410	0.149	+0.130
Biomedical	Source 50%	+0.261	+0.638	0.377	+0.336

Operational classifications provide additional description. Eight ACL conditions satisfy the joint mean rule; lexical-proxy-preferred 10% replacement is 4.59% [4.23, 4.95], below 5%. The source-preferred 10% lower bound lies only just above the 5%

criterion; the full-precision bound remains in the numerical resource, without implying a qualitative scientific divide at that threshold. All thirteen original rank-displacement lower bounds exceed 1%. Full estimates and assignment-variability summaries remain in Supplementary Tables S3–S4.

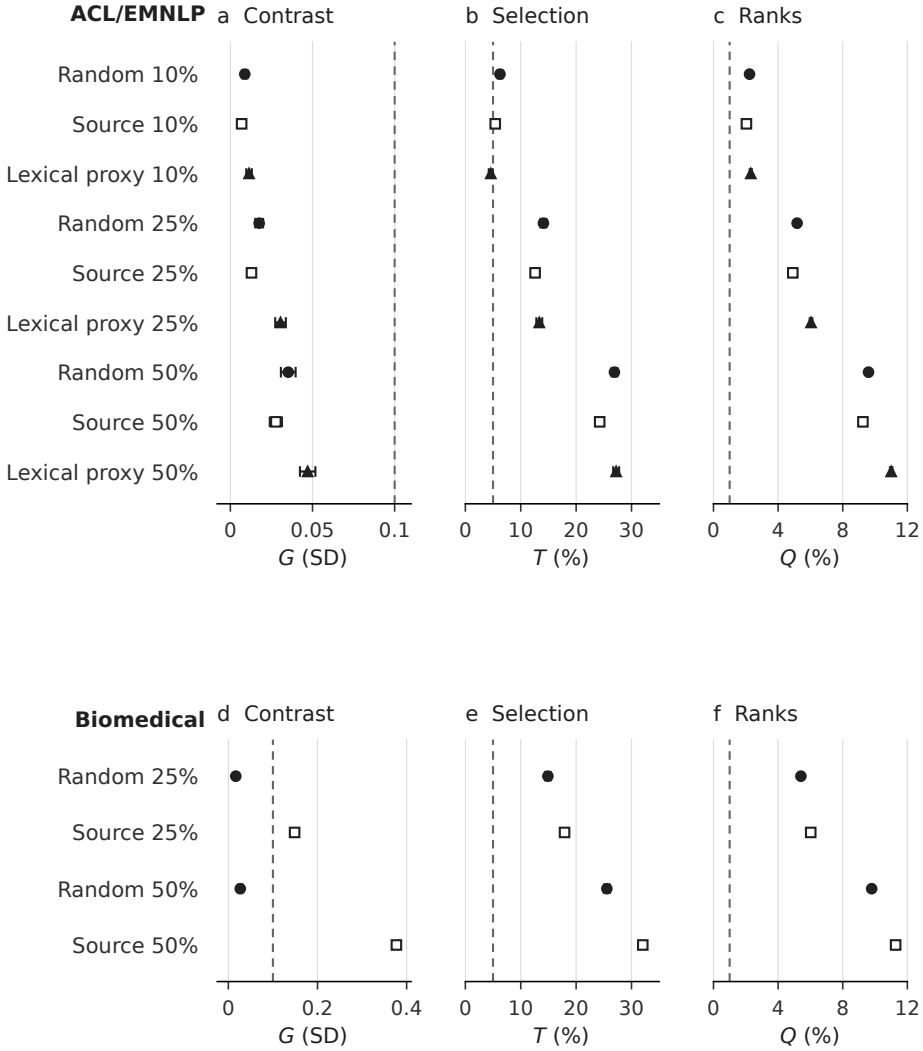


Fig. 2 Responses across measurement uses, with all thirteen original conditions. Points are means over 200 masks; whiskers retain separate approximate 95% simultaneous ACL (45-mean) and biomedical (18-mean) intervals. Dashed lines denote operating tolerances. G uses each collection's baseline score SD, T its original selected set ($k = 156, 148$), and Q its rank range (1,556; 1,470). Circles, open squares and triangles identify random, source-preferred and lexical-proxy masks. Panels a–c are ACL/EMNLP and d–f biomedical. The G axes use different explicit scales across collections; T and Q scales are shared. Narrow intervals may be hidden by markers

4.3 Structured versus matched random loss

Structured loss did not have a uniformly adverse effect (Fig. 3). ACL source preference reduced replacement relative to year-matched controls by 2.31 percentage points $[-3.00, -1.62]$ at 25% and 3.08 points $[-3.73, -2.42]$ at 50%. Its rank effects were negative at every rate; aggregate effects were negative at 25% and 50%, with the 10% direction unresolved. Biomedical source preference increased replacement by 2.89 points [2.21, 3.57] and 6.22 points [5.56, 6.88]. Its paired distortion effects were $+0.13154$ and $+0.34857$ SD, and all six biomedical paired intervals were positive.

ACL lexical preference separated the endpoints further. At 25% and 50% it increased distortion and rank displacement relative to year-by-source controls, but replacement decreased at 10% and 25%; its 50% replacement direction was unresolved. A condition can therefore cause substantial total replacement while causing less than its matched random control. These contrasts describe mechanism responses within each collection; matching does not explain their cross-collection difference.

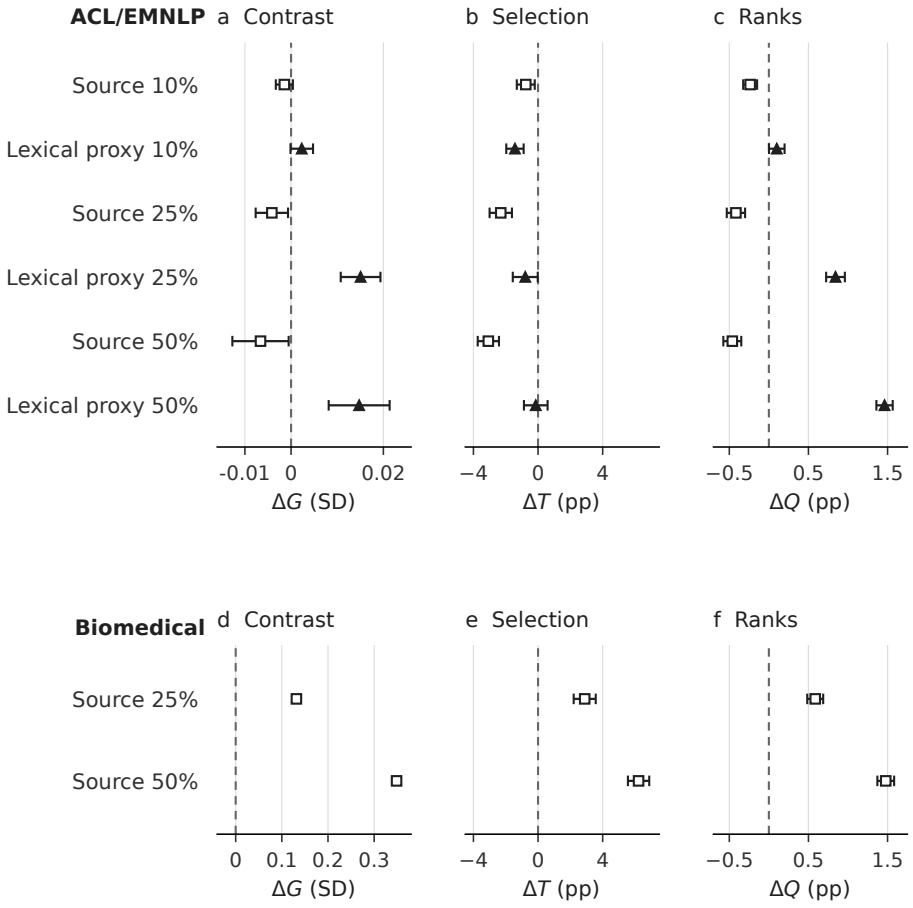


Fig. 3 Structured-minus-matched-random mean differences over 200 pairs, with original separate-family intervals. Source controls preserve historical-year missing counts; ACL lexical controls also preserve source within year. Negative effects indicate less distortion, replacement or rank displacement than the control, not an independently validated benefit. $\Delta T, \Delta Q$ are percentage points in the respective selection and rank-range units. The dashed reference is zero. The ΔG axes use different explicit scales for ACL (a) and biomedical (d); the other endpoint scales are shared. Full estimates appear in Supplementary Table S4

Descriptive neighbor accounting.

A post hoc diagnostic reused exact saved similarities for all original source-preferred 25%/50% assignments and their year-matched controls. At baseline, 93.3% of Computational Biology papers and 93.1% of Genetics papers had their nearest historical neighbor in the same source, compared with 42.7% of ACL papers and 55.1% of EMNLP papers. This describes stronger source separation in the biomedical nearest-neighbor structure, providing context for its unequal masking exposure without identifying a causal mechanism. In biomedical source-preferred 25% masks, the baseline nearest historical paper was masked for 38.2% of Computational Biology focal papers and 8.7% of Genetics focal papers; under controls the corresponding exposures

were 25.0% and 25.2%. At 50%, exposures were 73.4% and 21.2%. ACL source-preferred exposures were more balanced across focal sources: 23.0%/24.0% at 25% and 46.8%/48.2% at 50%. The fixed-denominator accounting reconstructs the source shifts (Supplementary Section S13.5). A paper can lose top-decile membership when competing papers' scores rise, so greater exposure of one source does not imply that the same source must lose more of its original selections.

4.4 Cutoff capacity and conditional selection loss

At ACL 10% loss, approximately 83–88% of expected lost selections began within the fixed 9th–11th-percentile band. The band contains only 16 baseline selections in each collection. If L_m is total loss and O_m is outside-band loss, then $O_m \geq \max(0, L_m - 16)$ for every assignment. Consequently, a positive outside-band count is partly forced when total loss exceeds this capacity. For random ACL 25% loss, the observed outside-band mean is 9.1 of 22.0 lost papers; the mean per-mask capacity bound is 6.0. At biomedical source-preferred 50% loss, the corresponding values are 35.2, 47.4 and 31.4 papers. The bound is accounting, not a null expectation or a measure of excess harm.

Figure 4 conditions loss on the number of baseline-selected papers in each fixed score-margin bin. Conditional loss probabilities are generally higher in the nearer-cutoff bins, although they are not uniformly monotone. Because most baseline selections occupy the final margin bin, that bin can nevertheless contribute a majority of losses at high masking rates. The smallest bins contain only one to five papers. Among papers at least 0.10 SD above the cutoff, mean loss at 50% masking is 18.6% under ACL random masks and 16.4% under ACL source masks, compared with 20.7% and 27.2% in the biomedical collection. These conditional probabilities describe changes within the observed selected populations; they are neither significance tests nor validated scientific costs.

The original count display remains in Supplementary Fig. S1, and the remaining fixed ACL conditions in Fig. S2. Rank-band and score-margin counts do not provide independent corroboration: at the existing 0.10-SD margin, the 132-paper positive-margin set is nested within the 140-paper outside-band set in ACL; the 132-paper outside-band set is nested within the 133-paper positive-margin set in the biomedical collection. Their losses must not be added. Supplementary Section S13.2 gives both capacity bounds and verifies that denominator-weighted conditional bin losses recover the original total losses.

4.5 Prespecified tolerance sensitivity

The contrast/selection separation at every ACL 25% condition and both biomedical random conditions remains supported with the stricter 0.05-SD distortion tolerance and 10% replacement criterion. None of the ACL 10% conditions meets the latter selection criterion. The ACL lexical 50% distortion interval crosses 0.05 SD; biomedical source-preferred 25% is unresolved at 0.15 SD, while its 50% distortion exceeds every tested aggregate tolerance. The full frozen grid is in Supplementary Table S6. These changes delimit operating classifications; they do not alter the continuous responses.

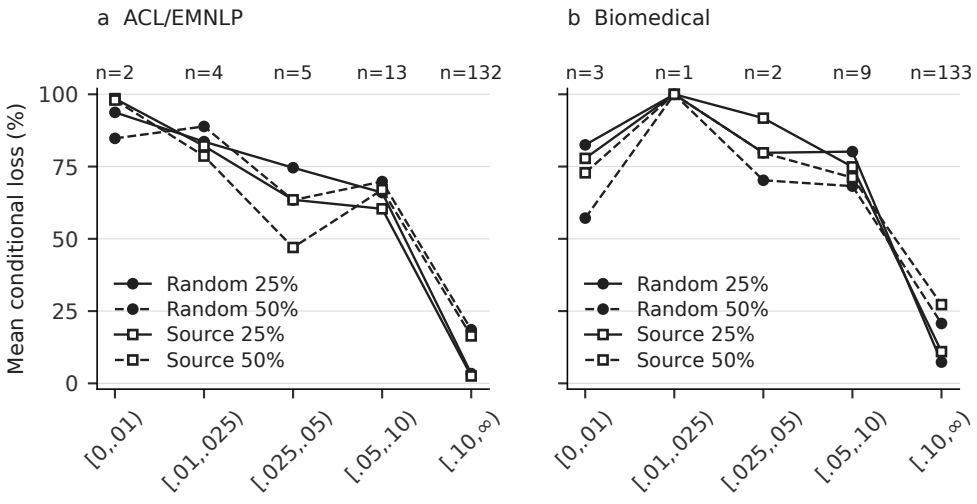


Fig. 4 Post hoc mean loss probability among baseline-selected papers in fixed signed score-margin bins. Each bin uses its own displayed baseline count n, and each curve averages 200 original masks. The horizontal axis is categorical: bin widths differ, with an open-ended final bin. Circles denote random masks and open squares source-preferred masks; solid and dashed lines denote 25% and 50% loss. Lines connect descriptive bin summaries and do not imply a continuous fitted response. No confidence intervals or significance tests are added. Empty bins, if any, are undefined, not zero

5 Post-confirmation robustness checks

5.1 Current-database coverage audit

A post hoc OpenAlex audit on 6 October 2026 found 7,176 uniquely matched historical DOIs with structurally valid, nonempty abstract indexes and fourteen ambiguous duplicate mappings among all 7,190 DOIs. There were no unique null, omitted or malformed indexes and no unmatched DOIs. The full-frame unique-mapping gate failed, and no informative known-omission pattern existed; no empirical-mask scoring followed. This dated coverage audit could not calibrate the synthetic loss rates and says nothing about earlier database visibility or broader database completeness (Supplementary Section S10).

5.2 Distributional sensitivity

A separately specified studentized bootstrap resampled whole masks and preserved structured/control pairs, using 200,000 fixed-seed draws per condition. Estimated intervals retained all 63 original mean classifications/paired directions and thirteen joint decisions. Finite-resampling uncertainty weakened one tiny ACL lexical 10% paired rank effect; the source 10% selection classification remained close to 5%. These results support the principal pattern against this distributional sensitivity without guaranteeing exact simultaneous coverage (Supplementary Section S11).

Table 3 Similar tested perturbation responses, different baseline comparisons.

A. Complete-history baselines

Collection	Encoder	Score SD	Contrast (raw)	Contrast/SD
ACL/EMNLP	SPECTER	0.027866	-0.003642	-0.130701
ACL/EMNLP	SPECTER2	0.012892	-0.001248	-0.096829
Biomedical	SPECTER	0.032666	+0.008535	+0.261287
Biomedical	SPECTER2	0.013068	-0.000100	-0.007625

B. Mean responses to 25% historical loss

Collection	Encoder	Mask	G (SD)	T (%)	Q (%)
ACL/EMNLP	SPECTER	Random	0.0175	14.10	5.18
ACL/EMNLP	SPECTER2	Random	0.0146	12.11	4.69
Biomedical	SPECTER	Random	0.0167	14.90	5.41
Biomedical	SPECTER2	Random	0.0170	17.23	5.39
Biomedical	SPECTER	Source	0.1486	17.91	6.02
Biomedical	SPECTER2	Source	0.1220	20.41	6.23

Each encoder defines its own SD, contrast, ranking and selected set. Source order is ACL minus EMNLP or Computational Biology minus Genetics. B shows means; original separate-family intervals appear in Fig. 2 and Supplementary Table S3, and SPECTER2 intervals in Supplementary Table S9. The SPECTER2 check is post-confirmation and reuses only these three conditions. No pooled scale or cross-encoder significance test is used.

5.3 Representation sensitivity

A limited post hoc SPECTER2 check (Singh et al., 2023) reused the original 200 masks for ACL random 25%, biomedical random 25% and biomedical source-preferred 25%, plus the 200 original biomedical matched controls. Each encoder supplied its own complete-history source contrast, SD, ranks and selected set. The nine base and three paired means used a separate approximate 12-mean interval family.

The tested perturbation pattern recurred (Table 3B). SPECTER2 random distortion was 0.0146 SD in ACL and 0.0170 SD in the biomedical collection, with replacement of 12.11% and 17.23% (18.9 and 25.5 papers). Biomedical source preference produced distortion of 0.1220 SD [0.1183, 0.1257], replacement of 20.41% (30.2 papers), and positive source-minus-control intervals for all three endpoints. All three rank-displacement intervals exceeded 1% (Supplementary Section S12).

The baseline comparison itself changed (Table 3A). Computational Biology minus Genetics was +0.261287 SPECTER baseline SD but −0.007625 SPECTER2 baseline SD: positive under the original representation, slightly negative and close to zero under the related model. Each standardized contrast uses its own positive SD, so the sign change also occurs in raw score units. The 0.10-SD operating tolerance exceeds the absolute SPECTER2 source contrast in both collections, so a below-tolerance classification cannot establish preservation of its direction. This is a comparison of observed baselines, not a cross-encoder significance test. Agreement about the response to historical title substitution does not validate the direction or magnitude of the original source comparison.

Complete-history rank and selection agreement also differed (Table 4). After exact identifier alignment, cross-encoder Spearman correlation was 0.684 for ACL and 0.589 for biomedical papers. Only 81 of 156 ACL and 62 of 148 biomedical top-decile papers were shared (51.9% and 41.9%); the overlap denominator is each encoder's equal-sized top-k set, not the union. Mean cross-encoder displacement was 17.36% and 20.24% of the rank range. Computational Biology represented 106/148 SPECTER selections and 97/148 SPECTER2 selections (71.6% and 65.5%); these are shares of the selected set, not within-source selection rates. These descriptive baseline comparisons concern one fixed cohort and do not identify a superior encoder. Source composition for both collections and truncation context are given in Supplementary Section S13.3.

Table 4 Post hoc complete-history agreement between SPECTER and SPECTER2. ρ_{cross} uses average ranks for exact ties; Q_{cross} is mean absolute rank difference divided by $n - 1$. Each encoder's top-k set uses its own scores and the original lexical-ID tie rule. These are descriptive baseline comparisons

Collection	ρ_{cross}	Q_{cross}	Overlap / k	Retained	Replaced
ACL	0.684	0.174	81/156	51.9%	48.1%
Biomedical	0.589	0.202	62/148	41.9%	58.1%

5.4 Title fallback versus record removal

In the separately specified post hoc comparison at 25% masking, removal produced higher mean top-decile replacement, rank displacement and cohort score drift than authentic-title fallback in all six collection-by-condition comparisons, including the matched controls. Under random masks, replacement was 14.10% versus 15.18% in ACL/EMNLP and 14.90% versus 18.02% in the biomedical collection (fallback versus removal). Under source-preferred masks, it was 12.59% versus 13.34% and 17.91% versus 22.76%, respectively. The original nonuniform matched-source direction persisted under both policies: source preference reduced replacement in ACL/EMNLP and increased it in the biomedical collection. The readouts do not establish uniform policy superiority: mean ACL contrast distortion was lower under removal in all three conditions. Full assignment and paired descriptive summaries are supplied for every selected readout. Supplementary Section S14 gives the design and compact table; Online Resource 2 supplies all selected conditions and readouts.

6 Discussion

The experiments show that the response of a semantic novelty indicator to incomplete historical text depends on the measurement use and the pipeline under which it is calculated. The principal result is a separation of outputs: random title substitution changed a source contrast little relative to between-paper score variation while altering paper ranks and high-score selections in both collections. The biomedical structured

intervention then supplied a boundary to that pattern, and the related-encoder check supplied a different boundary to interpreting the baseline comparison. These results support treating robustness as a relationship between a specified perturbation and a specified output, rather than a general property conferred on the indicator by one stable summary.

The contrast endpoint omits score-level drift. At random 25% loss, both source means rose in each collection, by approximately 0.10–0.12 baseline SD, while their difference changed much less. This observed cancellation is quantified by the drift/contrast identities; those identities do not identify a semantic mechanism. Under biomedical source preference, unequal shifts instead amplify an already positive contrast by 56.9% and 144.3%. Thus the absolute distortion endpoint should be read with the signed contrast and each group's shift. Aggregate/individual divergence has direct precedents (Donner, 2024; Liu & Zhai, 2026); the contribution here is the controlled, joint account of these distinct responses under retained-history title substitution.

Missingness structure also changes what can be transported. The biomedical random conditions reproduce the separation of source-contrast distortion from selection response, while source-preferred masking affects both. Source preference has negative paired selection effects in ACL/EMNLP and positive effects in the biomedical collection; lexical preference can move ranking and selection in different directions. These are informative limits of a universal worsening rule. Controls preserve specified opportunity counts while changing which documents lose abstract information. Their results concern the assigned information loss and associated content, rather than identifying a causal property of source labels. A random-loss audit therefore cannot stand in for an audit of structured loss merely because the overall unavailable fraction is the same.

Small contrast distortion can coexist with upward movement of both source means. Its apparent stability concerns a difference of means, not stability of score levels. The opposite matched selection effects show why a random-loss result cannot establish the direction of a structured-loss response in another collection. The same-mask comparison further shows that handling missing historical abstracts is consequential: retaining titles reduced mean turnover relative to removal here, while source-contrast distortion did not favor the same policy uniformly. This is conditional preservation of a defined baseline computation, not evidence that fallback yields truer novelty judgments.

Selection sensitivity also requires denominators and geometry. The fixed rank band can account for at most 16 losses per assignment; outside-band counts are therefore partly constrained by total turnover. The margin-bin curves report loss probabilities among the actual selected papers at each starting margin, with small denominators visible. Whole-cohort rank displacement supplies additional information about ordering, but nested rank-band and score-margin loss counts are not independent evidence of damage. None is a human-validated cost of changing the shortlist.

The SPECTER2 finding makes the pipeline qualification consequential. The complete-history biomedical contrast is positive under SPECTER, at $+0.261287$ of its baseline score SD, and close to zero but negative under SPECTER2, at -0.007625 of its own SD. Different positive scaling factors cannot account for the sign change. The low top-decile overlaps show that the disagreement extends to individual selection, even though the selected within-encoder masking patterns recur. Neither agreement

coefficient supplies a construct-validity benchmark. Within-encoder analysis measures how a defined calculation depends on its historical information. This operational question remains meaningful whether or not the calculation is a valid measure of originality. It does not establish the correct direction of the source difference. Agreement on conditional perturbation responses and agreement on what is being measured are separate evidential requirements; neither encoder validates the other, and pooling them would obscure this distinction.

The negative calibration and diagnostic findings clarify the practical reach of the evidence. The dated OpenAlex snapshot supplied no informative unique omission pattern, so the simulated rates could not be calibrated from it. The pilot coverage diagnostic did not improve held-out error over its simpler comparator. Consequently this study offers an evaluation design rather than a validated correction or completeness rule. The operating tolerances aid comparison, but continuous effects carry the scientific argument; the borderline ACL source 10% classification should not be made substantively special by a small threshold crossing.

For scientometric practice, the supported implication is to audit the output actually used and disclose the historical-information policy and representation that define it. A group-comparison audit leaves rank and selection dependence unanswered; a masking audit leaves construct and representation validity unanswered. Reporting the original contrast beside distortion, and the selection size and cutoff geometry beside replacement, exposes these distinctions. This combined evaluation adds a controlled measurement-use assessment to established reference-pool and metadata-quality research, with the second collection and limited representation check locating boundaries to broader interpretations of a small perturbation response.

6.1 Limitations

The experiments concern one nearest-prior score, authentic title-only fallback and two selected complete-text frames, with a bounded post hoc comparison against record removal at 25% loss. Their finite histories omit other scientific work, and current eligibility, exact deduplication and version handling restrict the represented populations. Synthetic loss rates and preferences describe conditional scenarios rather than estimated database omissions. Missing abstracts, corrupted text and deleted records are different interventions. The current coverage audit does not reconstruct historical availability.

Semantic distance is not independently verified scientific originality. Encoder training may contain later information despite strictly earlier comparison papers, and input truncation may affect representations. The limited related SPECTER2 check covers three conditions, not other encoder families or score constructions. The post hoc neighbor accounting describes different exposure patterns but does not identify a causal mechanism for the contrasting collection responses; source-correlated content, composition and truncation remain possible contributors. Matching fixes declared opportunity counts rather than source-label causation.

Intervals describe mask expectations conditional on each fixed collection, representation and assignment process. Approximate separate families exclude population, acquisition and model-selection uncertainty. Protocols were locally frozen sequentially,

and the second collection shares the first study's encoder and analytical construction. Neither the operational tolerances nor losses beyond a cutoff have externally validated scientific or policy costs. Scalar-to-endpoint reconstruction from the now-public package complements the prior saved-embedding checks; neither is a fresh acquisition and encoder reconstruction. The new diagnostics were specified with the original outcomes known and add no confirmatory inference. Final SemNovel methods remain uninspected, so overlap with the specific joint intervention is unresolved.

7 Conclusion

Within the two fixed complete-text reference frames, random historical title substitution produced small changes in a source contrast relative to between-paper score variation, but shifted overall score levels and altered paper ranks and high-score selections. Source-preferred biomedical substitution also amplified the signed source contrast. Similar tested masking responses across SPECTER and SPECTER2 did not imply interchangeable complete-history selections: 51.9% of the ACL/EMNLP top-decile sets and 41.9% of the biomedical sets were shared. Conditional loss probabilities and cutoff capacity clarify selection turnover without supplying validated costs of that turnover. The same-mask policy comparison found lower mean replacement under fallback in the tested conditions, without uniform superiority across measurement uses. These findings support auditing the score level, comparison, ranking or selection actually used, under the declared historical-information policy and representation. They do not establish scientific originality or general robustness outside the tested frames.

8 Statements and Declarations

Funding.

This study received no funding.

Competing interests.

The author has no relevant financial or non-financial interests to disclose.

Author contributions.

The author conceived the research idea, conducted the study, prepared the original manuscript draft, and reviewed and revised the manuscript. The author takes responsibility for the scientific content. This was an independent research project undertaken while the author was a student; no institutional funding or endorsement is claimed.

Ethics and consent.

The study analyzed published scholarly documents and public bibliographic metadata. It did not recruit human participants, conduct experiments on humans or animals, or access individual clinical records. No institutional ethics approval number or exemption is claimed.

Data availability.

Online Resource 2 contains the original and revision-stage numerical summaries and saved membership records, and a separately labelled policy-comparison folder containing the fixed specification, implementation, all 2,400 policy evaluations, complete descriptive summaries and verification receipts. Complete saved scalar and baseline scores, ordered identifiers, original masks, pairing mappings and independent endpoint-reconstruction code for the original experiments and bounded SPECTER2 analysis are publicly available in version 1.1.1 at https://doi.org/10.5281/zenodo.23298679 (Lin, 2026). Source texts, embeddings, model weights and detailed focal-neighbor records are not included. The removal comparison is not part of that earlier scalar package.

Code availability.

The public version 1.1.1 supports reconstruction of the original and bounded SPECTER2 endpoints from saved scores; the actual downloaded archive passed 5,000 arms, 75 family means and 116,898 assertions, plus 15 fixtures. It does not perform fresh source acquisition or encoder inference. The separate policy implementation is supplied in Online Resource 2 and the editable package; it requires saved similarities and project dependencies that are not distributed, so its original local verification is not claimed as a public scalar reconstruction. Supplementary Sections S8, S13.6 and S14 distinguish these scopes.

Supplementary information.

Online Resource 1 provides full methods, cohort accounting, all original and paired estimates, cutoff and tolerance diagnostics, the unsuccessful pilot, prior-art boundaries, post-confirmation checks and additional descriptive analyses. Online Resource 2 provides their numerical CSV/JSON tables with units, denominators and file hashes.

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